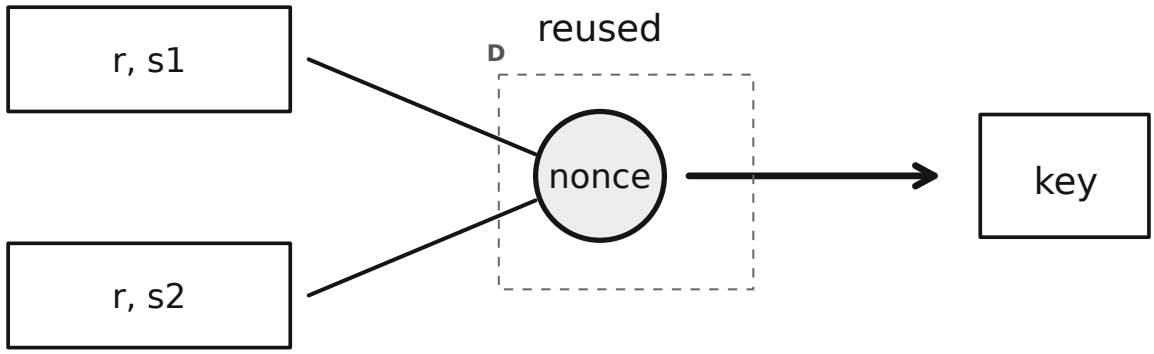
M134

ECDSA nonce-reuse recovery

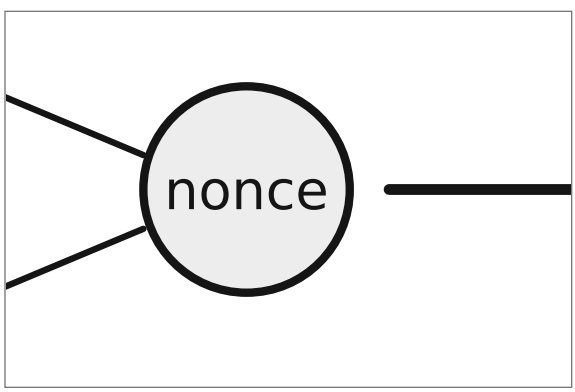


01 two signatures

02 same nonce / matching r

03 algebraic key recovery

D / OPERATION DETAIL



Vector detail / 1.60× main-view scale

HOW THE OPERATION WORKS

Reusing the same nonce for different messages creates two equations with the same unknown nonce, allowing elimination and scalar recovery.

READ THE THREE STAGES

01 two signatures
02 same nonce / matching r
03 algebraic key recovery

WHAT WAS ACTUALLY DONE

Synthetic sign/recover tests validate the algebra. Real harvested data are grouped by signer and r; the archived samples did not provide the required exploitable reuse pair.

ORIGINAL FINDING / SYNTHETIC/RECORDED TESTS

Algebraic recovery works in planted examples; harvested real signatures showed no matching reuse groups in recorded samples.